

E-Commerce Sales Analysis

Project Objective

The objective of this project is to clean up the data and analyze sales performance across product, category, customer segments and time period. The objective is to identify the causes of low sales and reduce potential losses in the future.

Dataset Description

1. Sales Data – which contains overall sales related information from Order date, Customer_Id, Product_Id, Store_Id, Quantity, Discount, Payment Type and Total Amount.
2. Customer Data – which contains Customer_Id, Name, City, Age, Gender, Loyalty_level
3. Product Data – which contains Product_Id, Product Name, Category, Sub-Category, Brand, Cost, Stock,
4. Store Data – Which contains Store_Id, Store Name, Region, City, Store Type

Column Description

Column Name	Description
Order_Date	Which tells when exactly sale is done
Customer_Id	Unique Id for each customer
Product_Id	Unique Id for each Product
Store_Id	Unique Id for each Store
Quantity	Tells the quantity sold each date
Unit_Price	Tells the unique price sold by each store for each customer based on ordered date
Discount	It tells discount percent of each store across customer for each product
Payment Type	Tells the mode of transaction

Total Sales	Tells the amount of the product sold after discount
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Data cleaning & transformation

Sheet Name: Sales_Data

Steps

1. Removed Sale_Id column as it is not required for analysis
2. Changed the Data Type for each column according to data
3. For Text column used TRIM, PROPER AND CLEAN functions
4. Date functions used for below column
 - Order_Date
5. Text function used for below column
 - Customer_Id
 - Product_Id
 - Prod_Category
 - Prod_SubCategory
 - Store_Id
 - Store_Type
 - Payment Type
6. Number Type used for below column
 - Quantity
 - Discount
7. Currency type used for below column
 - Unit price
 - Total amount

Formula Used

Sheet Name: Customer_Dim

Steps

J	K	L	M	N	O	P
	Customer Id inconsistent data is replaced					
	IF(LEFT(A2,2)="C-","CUST"&MID(A2,3,LEN(A2)),A2)					
	Using Find and Replace:					
	United states of America is replaced with USA. 3 Values has been replaced					
	Loyalty Level missing value is replaced					
	IF(ISBLANK(H2),"Bronze",H2)					
	Created a new column Age Group for Analysis purpose					
	IF(C2:C501<30,"Adult",IF(C2:C501<=50,"Mid Age","Senior"))					

	A	B	C	D	E
	Customer_Id ▾	Name ▾	Age ▾	Age Grou ▾	Gender ▾
	CUST1000	Sheena Bonilla	55	Senior	Male
	CUST1001	Eugene Nelson	26	Adult	Female
	CUST1002	Dr. Benjamin Price	29	Adult	Female
	CUST1003	Jeremy Harris	59	Senior	Female
	CUST1004	Caroline Collins	64	Senior	Female
	CUST1005	Christine Vaughn	24	Adult	Male
	CUST1006	Amber Jackson	29	Adult	Female
	CUST1007	Daniel Boyd	29	Adult	Female
0	CUST1008	Melissa Howard	53	Senior	Female
1	CUST1009	Ms. Maria Leach	64	Senior	Male
2	CUST1010	Dominic Page	53	Senior	Male
3	CUST1011	Brian Lopez	28	Adult	Female
4	CUST1012	Kenneth Lambert	55	Senior	Female
5	CUST1013	Nathan Hernandez	32	Mid Age	Female
6	CUST1014	Ashley Cohen	53	Senior	Female
7	CUST1015	Laura Alvarez	49	Mid Age	Female
8	CUST1016	John Scott	24	Adult	Female
9	CUST1017	Adam Cruz	60	Senior	Female
0	CUST1018	Rhonda Moss	43	Mid Age	Male
< > ... AaevsSales ReaionvsSales Customer_Dim Product Dim					

Sheet Name: Product_Dim

Steps

Cost Missing value is imputed using below formula									
IF(ISBLANK(F2),AVERAGEIF(\$D\$2:\$D\$101,D2,\$F\$2:\$F\$101),F2)									
Stock Missing value is replaced using below formula									
IF(ISBLANK(G2),AVERAGE(\$G\$2:\$G\$101),G2)									
Product Id inconsistency is replaced using below formula									
IF(LEFT(A2,2)="p-","PROD"&MID(A2,3,LEN(A2)),A2)									
Total Revenue generated by each product Created in Column H									
SUMIF(Sales_Data!\$E\$2:\$E\$2001,[@[Product_ID]],Sales_Data!\$N\$2:\$N\$2001)									
Total quantity sold by each product Created in Column I									
SUMIF(Sales_Data!\$E\$2:\$E\$2001,[@[Product_ID]],Sales_Data!\$J\$2:\$J\$2001)									

N14		Created in Column I							
	E	F	G	H	I				
1	Brand	Cost	Stock	Prod_Total_Sales	Prod_Total_Qty				
2	Hunt Inc	\$112.42	416	\$17,024.96	39				
3	Terry-Moore	\$342.56	181	\$17,226.44	41				
4	Sims Inc	\$296.83	352	\$15,827.99	41				
5	Pineda Ltd	\$364.67	452	\$24,311.58	47				
6	Gonzales-Davis	\$306.46	487	\$15,665.85	35				
7	Baldwin, Garza And Gonzalez	\$261.00	59	\$20,514.30	43				
8	Davidson Plc	\$312.80	350	\$9,937.09	33				
9	Harrell Llc	\$218.86	170	\$19,869.92	40				
10	Perez-Johnston	\$439.07	142	\$16,812.36	49				
11	Hill Plc	\$495.72	323	\$13,115.58	41				
12	Hamilton Group	\$298.86	308	\$18,539.25	51				
13	Gibson, Hobbs And Weaver	\$449.40	273	\$25,384.64	53				
14	Rice Llc	\$115.82	460	\$21,726.54	41				
15	Dawson-Perkins	\$242.23	453	\$20,960.03	55				
16	Luna, Anderson And Bowman	\$48.01	439	\$21,570.41	45				
17	Thomas Llc	\$465.34	301	\$28,285.61	72				
18	Erickson-Green	\$337.42	256	\$21,748.59	62				
19	Ford, Murphy And Wilson	\$144.53	367	\$28,324.05	70				
20	Guerra-Hendrix	\$303.75	343	\$15,084.94	37				

Sheet Name: Store_Dim

Steps

Used TRIM, CLEAN and PROPER function for each column

Sheet Name: Sales_Data

Steps

1.For Missing Values

Missing Values			
Quantity	IF(ISBLANK(H2),"1",H2)	5 values replaced	1
Unit Price	IF(ISBLANK(I2),AVERAGE(\$I\$2:\$I\$2001),I2)	3 values replaced	513.98
Discount	IF(ISBLANK(J2),MODE.SNGL(\$J\$2:\$J\$2001),J2)	4 values replaced	0.10
Total Amt	IF(ISBLANK(L2),H2*I2*(1-J2),L2)	4 values replaced	

2.To get data from another table

LookUp Functions used for below Scenarios:	
Using Product_Id fetched the Category from Product_Data table	
VLOOKUP(E2,Product_Dim!\$A\$1:\$G\$101,3,FALSE)	
Using Product_Id fetched the Sub Category from Product_Data table	
VLOOKUP(E2,Product_Dim!\$A\$1:\$G\$101,4,FALSE)	
Using Store_Id fetched the Store type from Store_Data table	
VLOOKUP(H2,Store_Dim!\$A\$1:\$E\$21,5,FALSE)	

3.Conditional Formatting Used:

- Highlighted Top 5 highest sales

	D	E	F	G	H	I	J	K	L
1	Prod_Categor	Prod_Sub_Categor	Store_ID	Store_Type	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
582	Sports	Equipment	STORE4	Online	4	\$968.27	0.03	PayPal	\$3,756.89
656	Clothing	Accessories	STORE3	Outlet	4	\$986.93	0.01	Credit Card	\$3,908.24
947	Home	Decor	STORE15	Flagship	4	\$973.90	0.02	COD	\$3,817.69
1241	Home	Storage	STORE9	Flagship	4	\$930.38	0.00	COD	\$3,721.52
1459	Electronics	Computer	STORE18	Flagship	4	\$975.47	0.04	Credit Card	\$3,745.80
2002									

- Used Icon sets For Unit_Price Column to identity high,low and moderate price of each sales

	D	E	F	G	H	I	J
	Prod_Categor	Prod_Sub_Categor	Store_ID	Store_Type	Quantity	Unit_Price	Discount
	Sports	Team Sports	STORE3	Outlet	1	\$218.04	0.09
	Electronics	Audio	STORE17	Flagship	3	\$104.47	0.27
	Beauty	Fragrance	STORE8	Online	4	\$914.05	0.25
	Electronics	Computer	STORE8	Online	1	\$921.51	0.16
	Beauty	Haircare	STORE5	Outlet	3	\$745.30	0.18
	Electronics	Appliances	STORE7	Online	3	\$147.01	0.19
	Home	Decor	STORE4	Online	2	\$785.31	0.15
	Home	Storage	STORE8	Online	4	\$342.39	0.23
	Home	Kitchen	STORE5	Outlet	1	\$448.25	0.07
	Home	Decor	STORE0	Flagship	2	\$873.17	0.23
	Sports	Outdoor	STORE14	Outlet	1	\$278.63	0.07
	Electronics	Mobile	STORE1	Flagship	4	\$286.92	0.17
	Sports	Accessories	STORE4	Online	4	\$644.58	0.01
	Electronics	Computer	STORE16	Flagship	2	\$316.90	0.30
	Sports	Outdoor	STORE15	Flagship	4	\$403.47	0.29
	Clothing	Tops	STORE9	Flagship	1	\$871.34	0.04
	Sports	Fitness	STORE19	Online	2	\$451.98	0.25
	Home	Decor	STORE4	Online	2	\$502.69	0.23
	Electronics	Mobile	STORE4	Online	2	\$559.23	0.25

< > Sales_Data Sales_Fact Dashboard Descriptive_Statistics ProductvsStockvsAmt ... + :

4. Date Modeling used

- Created relationship between all the 4 tables using the main table Sales_data

Status	Table	Related Lookup Table	
Active	Sales_Data (Customer_ID)	Customer_Data (Customer_Id)	New...
Active	Sales_Data (Product_ID)	Product_Data (Product_ID)	Auto-detect...
Active	Sales_Data (Store_ID)	Store_Data (Store_ID)	Edit...
			Activate
			Deactivate
			Delete

Visualization and Insights Arrived with screenshot of visualization Report/Dashboard screenshot:

5.Descriptive Statistics

Descriptive statistics is done for Total amount, Discount, Unit price and Quantity using toolpak

Steps: Data Tab → Click Data Analysis → Choose Descriptive statistics → Choose the input range → Choose output range and select summary statistics, confidence level for mean, kth largest and kth smallest → Click OK

A	B	C	D	E	F	G	H
Total_Amount	Statistics	Discount	Statistics	Unit Price	Statistics	Quantity	Statistics
Mean	1087.197349	Mean	0.14799	Mean	513.9847722	Mean	2.4765
Standard Error	18.3272372	Standard Error	0.001958251	Standard Error	6.307795342	Standard Error	0.024890433
Median	868.7182	Median	0.14	Median	522.43	Median	2
Mode	#N/A	Mode	0.1	Mode	513.9847722	Mode	2
Standard Deviation	819.6189642	Standard Deviation	0.087575651	Standard Deviation	282.0931835	Standard Deviation	1.113133992
Sample Variance	671775.2464	Sample Variance	0.007669495	Sample Variance	79576.56416	Sample Variance	1.239067284
Kurtosis	0.045826153	Kurtosis	-1.207282977	Kurtosis	-1.189706374	Kurtosis	-1.347725858
Skewness	0.87904704	Skewness	0.065097742	Skewness	-0.03676879	Skewness	0.029733624
Range	3889.7838	Range	0.3	Range	979.8	Range	3
Minimum	18.459	Minimum	0	Minimum	20.09	Minimum	1
Maximum	3908.2428	Maximum	0.3	Maximum	999.89	Maximum	4
Sum	2174394.698	Sum	295.98	Sum	1027969.544	Sum	4953
Count	2000	Count	2000	Count	2000	Count	2000
Largest(1)	3908.2428	Largest(1)	0.3	Largest(1)	999.89	Largest(1)	4
Smallest(1)	18.459	Smallest(1)	0	Smallest(1)	20.09	Smallest(1)	1
Confidence Level(95.0%)	35.94248725	Confidence Level(95.0%)	0.003840427	Confidence Level(95.0%)	12.37054179	Confidence Level(95.0%)	0.048813908

Pivot tables:

Table 1 : Product – Stock – Quantity – Total Amount

	A	B	C	D
1				
2				
3	Product	Stock	Sum of Quantity	Sum of Total_Amount
4	4K Home Projector	301	72	28285.6144
5	Adjustable Dumbbell Set	323	41	13115.5813
6	Adjustable Dumbbells 20Kg	423	45	20005.7362
7	Adjustable Jump Rope	316	28	14177.2243
8	Badminton Racket Set	358	53	23216.0621
9	Basketball	281	63	31070.4557
10	Beard Grooming Kit	256	62	21748.5882
11	Bedroom Table Lamp	329	45	19072.8195
12	Bluetooth Car Audio System	159	55	21716.8542
13	Boxing Hand Wraps	260	42	18013.0972
14	Brother Laser Printer	416	39	17024.9623
15	Car Dashboard Phone Holder	103	48	22759.2529
16	Carbon Fiber Tennis Racket	432	43	20744.2285
17	Ceramic Kitchen Bowl Set	308	51	18539.2507
18	Ceramic Plant Pot	273.4948454	36	16043.6629
19	Cricket Bat	471	55	24566.3395
20	Decorative Floor Vase	378	56	25205.8904
	< > ...	ProductvsStockvsAmt	YearvsAmt	CategoryvsQtyvsAmt
				Paym

Descriptive

- Product portable power bank gives the highest sales across other products
- Makeup brush set gives the lowest sales
- Men's casual sneakers has the highest quantity sold

- Vitamin C face serum is the lowest quantity sold

Diagnostic

- Power bank is used as a back up for our device hence it sold in high quantity and also yeilds highest sales.
- Power bank quantity sold is 84 and Men's causal sneakers quantity sold is 87 – both are almost equally sold and so both gives highest returns.
- Both products are widely used my many people.
- Makeup brush is not used by everyone. Hence it gives low sales and sold in low quantity
- Though Vitamin C Face serum has highest stock comparatively the quantity sold is very low hence gave low sales amount.
- Gaming mechanical keyboard sold in higher quantity and also gives good sales but it has low stock
- Wall mounted spice rack has lowest stock value but gives moderate sales and moderate quantity sold.

Table 2: Category : Quantity and Total Sales

	A	B	C
1			
2			
3	Category	Quantity	Total Sales
4	Beauty	738	\$321,561.33
5	Clothing	844	\$379,216.04
6	Electronics	1165	\$504,311.35
7	Home	965	\$427,295.80
8	Sports	1241	\$542,010.17
9	Grand Total	4953	\$2,174,394.70
0			
1			

Descriptive:

- Electronics category has highest quantity sold and also has highest revenue
- Beauty product has lowest quantity sold and also has low return

Diagnostic:

- Electronics product is used by both gender and also used across different age group. Hence it is sold in highest quantity and hence it gives highest revenue.
- Beauty products id used only by female and also certains age group female use these beauty products. Hence it is sold in lower quantity which gives low return

Table 3:Payment Mode and Sales

1			
2			
3	Payment Mode ▾	Sum of Total_Amount	
4	COD	\$702,459.42	
5	Credit Card	\$715,180.71	
5	PayPal	\$756,754.57	
7	Grand Total	\$2,174,394.70	
3			
9			
0			

Descriptive:

- Paypal has the highest revenue while COD has lowest revenue

Diagnostic:

- We could not see major difference in payment options.
- People use all 3 mode payment methods considerably.

Table 4: Gender, Age and Sales

	Gender ▾	Total sales	
	Female		
	⊕ Adult	\$328,885.76	
	⊕ Mid Age	\$542,528.92	
	⊕ Senior	\$360,029.55	
	Male		
	⊕ Adult	\$235,586.35	
	⊕ Mid Age	\$421,229.82	
	⊕ Senior	\$286,134.29	
	Grand Total	\$2,174,394.70	

Descriptive:

- Female mid age group has highest revenue
- Male adult has low revenue

Diagnostic:

- Female mid age group mostly investing in clothing and sports and electronics items. So that these age group people gives higher revenue compared to other age group.
- Male adult has less interest on clothing and beauty products. These group generate lowest sales
- Male mid age group in home category provides good revenue. Show that mid age male category was interest in home appliances.

Table 5: Regionwise sales

Region	Total Sales
⊕ East	\$622,404.32
⊕ North	\$646,100.88
⊕ South	\$635,213.10
⊕ West	\$270,676.41
Grand Total	\$2,174,394.70

Descriptive:

- North Region gives highest revenue while west region gives low revenue

Diagnostic:

- Only 2 stores were operating in west region. It might be the reason it gives low revenue
- Other 3 region has 7 stores equally operating. Hence all the 3 region gives almost equal sales revenue and north has the highest revenue while south and east region has good revenue generating.

Table 6: Yearwise Sales

Year	Total Sales
2023	
Qtr4	\$223,812.72
2024	
Qtr1	\$268,596.98
Qtr2	\$245,299.96
Qtr3	\$285,586.04
Qtr4	\$294,428.82
2025	
Qtr1	\$271,630.62
Qtr2	\$256,716.06
Qtr3	\$289,855.68
Qtr4	\$38,467.82
Grand Total	\$2,174,394.70

Descriptive:

- 2024 has the highest revenue while 2023 has lowest revenue generated

Diagnostic:

- Only 4th quarter is considered in 2023 hence it shows as lower performed year.

- While you look closer in 2025 4th quarter is not covering all the months hence it shows lower sales.

Business Questions answered:

1. Sales Analysis:

- What are the monthly/yearly sales trends?
- Which product(Category/sub-category) gives the highest sales?
- Which payment method is used most widely?

2. Customer Segment Analysis:

- Which age group and gender contribute more to sales?
- Which gender generate highest sales?

3. Product Analysis:

- Which product is best selling?
- Which products have high sales but low stock?
- Which products have low sales?
- Which product has the highest quantity sold?
- Which product has highest stock but sold in low quantity?

4. Store Analysis:

- Which stores generate the most revenue?
- Which regions perform best?

Predictive Analysis:

Products like 4K home projector, gaming mechanical keyboard, Laptop cooling pad, men's casual sneaker, vitamin E body lotion, wireless mouse gives good to moderate sales.

Prescriptive Analysis:

Vitamin C Face Serum has low quantity sold but has more stock. Need to provide some discount to sold out the remaining stock.

Bringing more stock to the product Gaming mechanical keyboard will give a good sales

Products like Kitchen Cutting Board, Knee Support Brace, Men'S Leather Belt, Multi-Purpose Storage Box, Portable Bluetooth Speaker, Wall-Mounted Spice Rack, Women'S Ankle Boots has low stock. Increasing the stock of these products provide good sales.

Product make up brush set, Men'S Formal Shirt, Herbal Face Moisturizer, Women'S Yoga Pants has high stock compared to quantity sold.

Rugby Ball, Stainless Steel Cutlery Set for these products we need to further investigate the reasons for low sales in terms of quantity sold and stock.

We can increase number of stores in west region to provide good sales.

We can bring more number of women's clothing brand and provide more discounts to yield good sales.